

ActInf Livestream #007.2: "Variational ecology and the physics of sentient

Livestream | 1:12:36

hello and welcome everyone
to the active inference live stream this
is
active inference live stream 7.2 it is
november third
twenty twenty welcome to team com
everyone we are an experiment in online
team communication
learning and practice related to active
inference you can find us on twitter at
inferenceactive
at email our public key base team or on
youtube
this is a recorded and an archived live
stream so please provide us feedback so
that we can improve on our work
all backgrounds and perspectives are
welcome here
and also for video etiquette on live
stream remember to mute if it's noisy
raise your hands so we can hear from
everyone in the stack and use respectful
speech
today in act imp stream 7.2
we are going to go through our
introductions and have some warm-up
questions
and then we'll get into the discussion
of 7.2
we'll talk about the paper variational
ecology and the physics of sentient
systems
ramstead it all 2019 and we'll go over
the goals and the abstract
and the roadmap then we'll talk about

some big ideas and questions in ecology
and then we'll go through the figures i
remember last time we mentioned figure
six and so we've gone
the figures in reverse order so we'll
start with six and we'll just go down
from there

and also for the rest of 2020 we will be
discussing papers
8 9 10 and 11 which we'll go through in
a second

and just check out our twitter or
message us if you have any questions
so what are papers 8 9 10 and 11 and the
dates

paper 8 scaling active inference on
november 10th and 17th

all discussions are at uh 7 30 to
9 00 pst pacific standard time

in the morning paper 9 is
projective consciousness model and
phenomenal selfhood

paper 10 is a variational approach to
scripts and

paper 11 is sophisticated active
inference

which as we were just talking about
before the stream we'll also have
some cool accessory content let's go
to our intros and warm-ups so for this
part just introduce yourself
and your location you can say hello and
give a short introduction and then pass
to someone who hasn't spoken yet
so i'm daniel i'm in davis california
and i'll pass it to shannon
hi guys i'm shannon i'm usually in
merced california but i'm

in south dakota for the pandemic and
i'm going to pass it to ivan
hi everyone my name is ivan i'm from
moscow russia
and i pass it to sasha
hi everyone um i'm sasha and i'm in
california
and i'll pass it to marcus
hi i'm marcus um i am in
lean shopping which is a small town in
sweden a couple of hours outside of
stockholm
um and i gave a brief introduction to
myself because i'm new here
so my i did my phd in medical imaging
and my research is mainly focusing on
cardiovascular mri so quite different
from fap
and active inference which has become
sort of a hobby for me
the last couple years and i find these
live streams very good to explain
these concepts and i'm looking forward
to the discussion today so i'm gonna
pass over to cameron
hi i'm cameron i'm in zurich
i was here for the first time last week
and i
just started a phd in philosophy in
zurich
uh and i passed it on to
alex i'm alex
i'm in moscow russia i'm a researcher in
system management school
and i pass it to maxwell
right thank you hello everyone uh my
name is maxwell ramstead i'm a
postdoctoral fellow

at mcgill university and i'm zooming in
from

just outside montreal in the suburbs
and i'm a first author on this paper so
very excited to

discuss it with you

i i'm not has everyone spoken yet i

i lost track okay

great thanks for all the introductions

let's go to the warm-up questions

and here it'd be cool to hear from

everyone and you can just

answer as narrowly or as broadly as you
like

and go ahead and raise your hand if

you'd like to speak here

the first question is what is your niche

or what could it be so i'll just start

with my local

niche right now i'm looking at the video
screens i'm

trying to balance getting the

information from people's face versus
the

slides um what's another person's just

niche right now around them

or a related question is what skills

help us find affordances in a changing
world

so our niche definitely presents us with
uncertainty

multiple scales perhaps of uncertainty

what's behind me in the door or what

happens in the deep future

so how do we find affordances or how do

organisms find affordances in that kind
of a context

i liked your mention of your niche is

the computer screen and looking at
people's faces

i feel like for every different
communication program like if it's zoom
or jitsi

we can organize our digital environment
different ways so we can like see
different people's faces
or see different aspects of the screen
to allow for as much
non-verbal communication as we want or
need

to make that particular communication
effective

cameron um

for the first question um going back to
a niche i constructed for myself
about a year ago when i was coaching
tennis would often work in the
late afternoon to late in the evening
and i noticed how my diet changed to
make sure i had enough high enough sugar
level

so that i could give really energetic
lessons to the kids

in the late afternoon and then
sort of tone it down in terms of an
energy level for the adults
and then uh have a massive kind of slump
as soon as i got home so i could sleep
properly for the next

kind of round of coaching so i noticed
how my kind of diet

set up my energy levels just for that
kind of lifestyle and as soon as i
stopped coaching tennis

it was uh away with the biscuits and
chocolate

yep that kind of just reminds me of like
looking forward to dinner you're not
exactly sure what it's going to be but
it's going to be a physiological thing
it's going to reduce your uncertainty
about food
anyways what is one thing that you are
curious about or you want to explore in
today's discussion
so that could be something that you
wanted to figure out from the paper or
something that the paper
kind of resonated with you or just
something
from the side from the rest of your life
that came into context
today when thinking about this paper
so i'll start one thing that kind of
motivated me in the slides was just
exploring how the variational ecology is
linked to
big questions in so-called i guess
traditional ecology
non-fep non-active inference ecology how
do we
make that connection because if that
connection can be made
then the expansion of the theory from
just a few equations to something that's
in use in a lot of
applied situations would be a lot more
easy
any other random thoughts or that's
totally
chill we can just go on to the paper and
people can raise their thoughts
as they see fit i mean i'd be curious to
hear

uh like everyone else on this like what
did you hope to get from this
uh because as the author i can respond
to some of these questions so
let's go cameron and then shannon
muted sorry
at the very end uh maxwell you
you write about shared intentionality
and i wondered how
you think about the debate between
shared intentionality and
theories of mind and
that kind of kind of very mentalistic
idea of interaction
yeah i had that sort of same topic of
shared intentionality
i was also wondering um in our
universities like seminary yesterday we
were talking about collective memory
and i wonder whether this
like variational ecology can
tell us anything about what we choose to
remember or what we
end up remembering as a culture or as a
society about
our culture's past it might not
it might be that might be a little
abstract for what we're talking about
currently since it's not necessarily
behavior or it's a behavior
that we have to verbalize and speak out
loud but
those are great questions yep
i agree on the part about remembering
it's like there's this multi-level
remembering process how much data do we
retain from yesterday from
two seconds ago from 200 years ago from

2 million years ago
and then there's all always different
perspectives on just the
facts or the events of whatever did or
didn't occur and all this narrative
weaving that occurs so how we
model and remodel how we update our
models of those
multi-scale memory processes some of
them are extended some are internal
that's everything so i think it's really
interesting question
i hope we'll return to that sasha
um yeah that's a kind of great point to
link it all together that
um individuals that are interacting
um can spur more i guess memories or
common experiences
by revealing more or less about their
current niche
um and so yeah just really thinking
about how
individuals should um or should not
interact
um when trying to build common ground um
and how much of their niche they should
be sharing uh in that communication
but yeah a lot of uh fun things with
that in the virtual space
um that we can reveal to each other
cool i think that hopefully sets the
stage to talk about
some of the specifics of the paper and
also big ideas in ecology
and also things are happening all around
us so the paper is
variational ecology in the physics of
sentient systems from 2019

and we reviewed the goal of the paper
last week
but the shorthand kind of equation form
is that
variational neuroethology vne which is
about
organisms and their neurobiology
primarily
can be kind of jointed together with
vanc
the variational approach to niche
construction and then
that summation that continuation of
explanation from kind of the
neurobiology
into the extended niche is called
variational ecology
and then the non-active inference
phrasing would be
something like how do you make a
multi-scale ecology framework that
builds on
ecology evolution and development and
also adds in insight from physics
mathematics and complexity
we also talked about the roadmap
last week and sort of formulaic in that
it introduces the two separate ideas
in section two and three and then adds
them together
to result in a synthesis framework
represented here and with the connector
added between the uh
active orange connector to the purple so
to kind of just start the discussion i
wanted to begin with some big questions
in ecology
and some other people may have done

ecology

in a specific formal or informal way but

there's so much to it

and it's just interesting to connect it

to the questions that motivated

that field before this kind of synthesis

of

even ecology evolution and development

which is its own pyramid of

you know ideas that connected and people

who bridged evolution and ecology and

all these kinds of things so let's just

look at the big questions in ecology

and think about the context that this

synthesis is happening in

and then think about all the cases where

ecology is applied so conservation and

policy planning and stuff and then think

about how a mapping between

fep and active inference broadly and

ecology

could allow for a lot of translational

and applied use

so here's one kind of interesting

historical coincidence

it's a 1983 paper by richard lewinton

called the organism as the subject and

object of evolution

so this is kind of before the whole uh

physics in biology approach was becoming

mainstreamed but he was coming from a

very

integrationist perspective and he wrote

before darwin

theories of historical change were all

transformational that is systems were

seen as undergoing change in time

because each element in the system

underwent an individual transformation
during its life history so like the
muscle cells became different or
something

lamarck's theory of evolution was
transformational in regarding species as
changing because each individual
organism

underwent the same change through inner
will and striving an organism would
change its nature and that change would
be transmitted to offspring so that's
what people often call the marquee in
evolution

now here's where the historical
coincidence is

in contrast darwin proposed a
variational principle now it turns out
that

this is a different variational
principle than the variational principle
from physics that we're going to be
discussing but i thought it was really
interesting that lewinton and darwin
use the same idea and so what's the idea
individual members of the ensemble
differ from each other in some
properties

and that the system evolves by changes
in the proportion of different types
there's a sorting out process dot
dot and so variation among objects is
transformed into temporal variation
a dynamic process arises in time as the
consequence of static variation in space
there's no historical process other than
the evolution of living organisms that
has this variational form at least as

far as we know
so that's actually going a long ways
towards this sort of what is
schrodinger's
question about what is life just an
interesting
way that variational has interwoven from
the history
of biology from darwin from lewinton and
now finding itself with a formal
definition
ironically with the same name related to
variational principles so anyways
sorry for the little bit of a
uh history side let's get to the ecology
questions and if anyone has thoughts on
that previous link they can kind of
address it here
these are some questions drawn from a
1999
paper by sir robert may called
unanswered questions in ecology
so we can figure that they're probably
still unanswered but
let's just look at some of these and if
anyone wants to reflect on it
how do they see it playing out or how
do they think or wonder that active
inference could play into it
i'll put up the first two what
determines population density
and what determines the spatial
structure of population these are
actually like the first two
that may brings up so what what
could an answer to this look like
uh or where do you think
the fep and ecology could intersect here

i'll just list them all also how can we
study stability and complexity in
ecological
communities and what is the proper scale
of analysis for ecological studies in
space and time
so what might be an approach or what
might be a mapping
that we could draw here
one thought is the proper scale of
analysis for ecological studies
is kind of related to our discussion
about what time scales
ergotic systems or the ergotic
hypothesis
exists at because there might be some
time scales or spatial scales that don't
make sense to study
like if you're studying the lion hunting
the gazelle
the centimeter squared time scale or
spatial scale
in the one second might not capture some
important dynamics of the system
but then if you study it at too broad a
level it might be
not able to be studied for another way
just happening too fast underlying
so just one thought there
anyone wanna yeah shannon go ahead
yeah so i don't study ecology but um i
know we've learned some about
like predator prey models so population
density
is determined by you know the amount of
resources in the environment
and as the resource depleted then the
predators depleted because they don't

have enough food so the prey can come back and you have this structure over time and the same kind of thing happens even with trees and these beetles that eat trees so you'll see like forests even over a much longer time scale having um like trees that die out and then they grow back eventually and the beetle population will also follow that predator prey structure and die out and come back yes so there's these rhythms and some slower ones some are across species some are within a species they're all related to measurements so at the very least we're going to be able to use different mathematical frameworks to analyze our measurement just like we use different statistical tests but the deeper question is whether the top or whether the active inference framework describes what these ecological communities are doing or what is actually occurring versus to what extent it's a really convenient model fitting approach um all right the second next slide yeah what would the difference in implication be i mean uh i think uh the proper philosophy of science attitude to have with regards to any scientific model whatsoever is that truth has nothing to do with

what science is doing i mean this might be slightly controversial but i think science has to do with uh model comparison effectively i mean all of science has moved into this kind of kind of game i think where what we're doing is so first of all data is never presented in a naked way right uh so you know your neuroimaging data for example will be organized as a kind of a heat map of the brain for example like data is always presented as a model of data and basically what we're doing is confronting different theoretical and formal models to find which explains uh away the variance in our data the best so from that point of view we needn't read the free energy principle as making any literal claim about anything just the same way that you know uh any other scientific model is just going to be kind of a it's a useful description so i'm i'm putting on my mel andrews hat today and being an anti-realist but uh or i mean a non-realist right like it's just uh it's not like newton's laws became false after you know they were shown to be a special case of relativistic uh you know dynamics these are just like successively better models that successively and iteratively explain away the variance in our data

better

but i don't know if it's appropriate to

ask the question you know

are these merely models or actually the

way that the world works i mean

i mean that's the quote right all models

are wrong but some are useful

exactly yep and then the what is useful

must always be unpacked to to whom

and who else is influenced by it and

then what is their value structure and

all these types of second order effects

and and i

agree that if nothing else it's a useful

multi-scale framework just like a

multi-level

regressor regressor model or the

gaussian distribution i mean linear

modeling is used for some computational

benefits so

yeah it's unfortunate when it does

constrain the

phenotypic distributions in the world

but perhaps with flexible tools we could

actually accommodate for

very very high dimensional preference

systems or things like that

rather than using like a excel

spreadsheet that

takes an average and then aggregates

that into the next level of the

aggregator

so how could we go and look at variation

across multiple levels of the system and

yeah good interesting things there

let's look at a few more just classic

ecology questions before we

go to the figures so this is from a

2013 paper in journal of ecology called
identification of 100 fundamental
ecological questions
so those are just the little small ones
um i'm just going to list
all 20 of them that i just grabbed the
first 20
um so people could can uh think just you
know pick a random number
spin something roll roll three die
whatever it is
but these were just the questions that
this large set of authors basically
used some polling and discussion
methodology
to converge on these 100 questions and
so they definitely thought about which
ones to put
at the beginning and they definitely
thought about which ones to include so i
just think these are kind of
what ecologists broadly may
find to be some of the biggest questions
so that means
that if somebody can make a connection
where something that was previously
modeled in an incomplete or
in a really hetero heterogeneous
framework
if there's a way to model it in a more
unified way
then that's something that addresses a
fundamental question in ecology
and so that's one of the ways that we
can apply theory
and then think about what are the
downstream applications like look at
their number one question

what are the evolutionary consequences
of species becoming less connected
through fragmentation or more connected
through globalization
so that's roads being built that break
uh migration corridors
that's also barnacles being brought
around by global
fishing fleets and so different species
and different
ecosystems are connected in new ways
their markov blankets their ergodic
levels of analysis these are all
changing and so
how is that going to shape phenotypic
and behavioral and genetic diversity
that's conservation ecology so
if that can be re-understood or
mapped into a active inference framework
it's again at the very least a super
useful way to think about multi-scale
systems
because it includes that bottom-up and
top-down
element which a lot of these questions
implicitly include
but often through very constrained ways
like
uh specific types of partial
differential equations or
specific types of simulations that
aren't really grounded in this type of
methodology
and then um yeah i just think that's
kind of an interesting
thing so i thought that's that's why
it's worth taking the time to
at least look through some of these

anyone can raise their hand if they want
to jump in on one of these
1 through 20 but like it's just useful
to think about these because these are
the questions that people want to see
answered in ecology shannon
i mean number five is exactly what we're
addressing if we're asking what can
multi-scale
uh free energy tell us about
these different levels of of selection
and also different levels of behavior
or different levels of insert phenomena
you're interested in here
yep what are the relative contributions
of different levels of selection gene
individual group
it's a life history evolution and the
resulting population dynamics
yep definitely something that's super
interesting to think about
and does a variational principle that
links these levels of analysis like if
you have a multi-level regression
model you don't say that one level has
causal priority over the other level
it's just a different way that variance
was partitioned
and that reminds me of evelyn fox
keller's metaphor
of the two people who are pumping water
together
and that's the nature nurture so-called
and that's
how you can uh really do a partitioning
of labor only for certain tasks but not
for kind of shared tasks
like her water pumping example and so

maybe the partitioning is a statistical
post hoc and
some traits will partition highly on to
heritability or some traits partition
highly onto some other
uh uni-dimensional axes but then in a
fuller variational framework
we would have a more nuanced way to talk
about how those parts are related
yep and then another question that is
definitely related to the whole area is
16.

how do organisms make movement decisions
in relation to dispersal
migration foraging or mate search so
definitely a lot of
work on everything from the ocular motor
epistemic foraging
the movement of the eyes towards
informative areas of the visual space to
focus on
to search behavior and not could be in
an
abstract computational space so many of
the optimization algorithms we know
are named after animals because they're
kind of bio-inspired ant colony
optimization
cuckoo search all these kinds of things
are related to
ecologically observed strategies so
maybe if we had a way to frame
those different strategies in nature we
could do bio discovery and then we could
implement a lot of cool ways to who
knows what you know
optimize things make uh swarms of robots
that are useful

who knows what should happen okay um

number eleven

also uh that what is the what are the
evolutionary and ecological mechanisms
that govern species range margins
um that reminds me of this uh concept in
ecology of uh the fundamental versus
realized niche

um and so i think that has
a lot of uh nice parallels to um
kind of how humans behave but also
thinking about um
cognition and how we
uh choose to think about the world and
what sort of niches that
uh that model is tailored to
what is the fundamental versus the
realized niche

um uh full disclosure i also do not
study ecology but i'm teaching an intro
to bio class
to freshmen um and so i'm learning about
this

uh as i go as well it's it's kind of fun
um but
the fundamental is the
environment uh the combination of bio
abiotic factors

that allow an organism to survive so
anywhere where you could live
um and then the realized niche is uh
where you actually do
live based on your interactions with
other species and other
biotic factors that influence your
niche do you know what this overlaps
with corolla

thoughts's concepts of the developmental

versus the selective niche
so as thoughts basically argues that
effectively uh you can think of the
niche
as the set of selection pressures to
which organisms are subject
but also as the set of reliably
inherited resources that are necessary
for the reconstitution of the life cycle
in every generation and that in
ideally they coincide but they don't
necessarily
uh so that you know your developmental
niche
ideally prepares you for the pressures
that you'll be you know undergoing
in your selective niche uh but you know
for various you know evolutionary
reasons or other
you might be tuned to the wrong
environment for example and then die off
so
uh but it does seem to overlap with the
distinction that you're mentioning here
wow very interesting and
the changes in the uh in different
things
can lead to the realized niche where
things actually exist
if the population hits a critical point
then it can just
extinct locally or or even regionally
and that doesn't that's like a non-uh
recoverable state
and so it can change uh
in a very like the niche can be modified
rapidly
and so that definitely relates to it

let's actually do one more slide on
niche and affordances
and then this is before we just get to
the figure so if anyone has any more
like general
thoughts now we're bridging the gap
we're
seeing how the regular ecology brings
over to this new way of thinking about
it
so niche and affordance are linked and
that's kind of cool that we just were
talking about that
because there's the niche example here
is the different birds so flamingo is
digging the deepest into the dirt
to or under the water to get food the
um birds on the right are going more
shallowly and so their beaks are
different their behavior is different
everything is different about their
biology and
they may or may not be able to have uh
general versus specialty
like maybe the flamingo can also get the
shallow but not vice versa or something
like that
so that's a sort of physical niche
concept but our niche
includes objects includes other people
and trees and
bike paths and digital experiences as
well
so what happens now that technology is
part of the human
informational and ecological niche maybe
bringing in what was just being
discussed about the fundamental

and the realized as well as the
selective and the developmental
niche concepts so how could active
inference be used to
to understand and model and design in
these situations
so i gotta start there is for politics
or
for information systems especially with
digital
there is certainly a vast number of uh
degrees of freedom in the design and the
realized info niche that we're in is
just like literally whichever whichever
one we're in
however it ends up whoever's behind the
scenes doing whatever that's the
realized one we're gonna get
but what is the fundamental info niche
what is the actual
bigger habitable space and which parts
of that habitable space might have
some features or some benefits that we
don't
realize through our current uh
existence in that realizable so
conceptually i would want to distinguish
between the physical environment and the
niche
uh so the way that we're defining it at
least the niche is the set of
affordances to which
organisms are sensitive so that doesn't
overlap
exactly with the physical environment um
so
i mean two species can share the same
physical environment

but not occupy the same niche because
they're not sensitive to the same things
uh so you know like something may be
like a prey item for one
and just be like you know part of the
background
uh for another species uh
so i think that's how you get around
that uh i was confronted by kim sterily
uh when we we talked we gave a talk in
australia
i think in 2017 or 18
about this and i think this helps to
dissolve that issue there isn't like one
ecological niche there's one physical
environment that we all share
and then relative to uh essentially the
the skills and the concerns that we
embody
uh the world will appear very
differently and
so we we live in different niches even
if the physical environment overlaps
okay yvonne
yeah thank you if if we return that
now we share one virtual
niche uh we do it by
uh our brain work and
before the internet we can
uh speak with each other
just to physically present
near or by telephone but now
we can we can do it
we can choose what type of communication
we can choose the type of communication
and
so we construct
different ways to

create niche itself
yes thanks for sharing that
and it's uh i i hear you max well
on the niche concept definitely an idea
that ecologists have debated over
as much as um species
but the uh idea of the shared
informational resource
let's go with that certainly there are
things that are becoming knowledge
resources that are
not only uh non-evenly shared across the
globe but also are related to new kinds
of informational connections that
haven't occurred
in any individual sensory niche
hearing languages that one wouldn't have
heard access to perspectives and
the high-throughput nature of media so
these are all
things that i think change our our
match or mismatch or our fitness between
our evolutionary priors on biofeedback
or on safety or on the
veracity of language these kinds of
things
are being connected and being
played out in new ways so i would think
how can active inference be used to
understand and design these
situations well we can define the remote
team
and we can also think of it as almost a
reduced
case and a known mismatch case with our
own
like quote natural ecology but first our
ecology is so flexible

and humans have displayed so many lifestyles that this is really one of many there's no reference wild type that's one thing and then the second thing is when you take an animal into a laboratory you know that it's not going to be getting the same microbiome or the sun or the temperature variability or the prey or the predator or whatever it is but you you go you forego that degree of freedom and naturalism for an element of controllability and so in ecology they do a lot of mesoscale experiments where they'll set up like 200 fish tanks and add exactly three salmon and exactly 11 algae and set up these tanks and they can demonstrate things like dynamic oscillatory systems or like a instability point with the ratios of different initial conditions of two different species or the role of different nutrients and then the the critique just like it is with animal behavior in the laboratory is like well that's not exactly the natural ecosystem and so it's just an interesting topic how in ecology there's this continuum from super long scale systems like cosmology that you're never going to be able to uh intervene in or make experiments about like tectonic plate level

understandings of biodiversity ranging
to very local
understandings that are often very
controlled but then
those are often subject to the same
kinds of holistic criticisms that
so many other frameworks are and so to
turn again to
active inference there's a way to build
positively
on this and just think great there's
this mismatch between levels or at least
apparently
why don't we kind of think from the top
down and set out a really
big scaffold for multi-level systems and
then we'll figure out how to
map different levels and scales of
analysis within this just like people
learned how to use multi-level
regression
this is going to be like a variational
tool at that kind of category or
of that kind of type all right
figure six so here maxwell on figure six
i'll let you take a first pass but the
next slide if you
just you let me know if you want the box
to come up but let's just focus on this
uh image and maybe start off just like
what
does it convey or what is it there for
but go ahead
okay uh just let me pull up the
the figure myself all right
um it's called a particular partition
it's a point
yeah just to explain it i know that i

just wanted in front of me while i
explained
well so basically this this figure
um explains how
uh you add i mean i i would
i would prefer to explain things in
terms of the
figure figure 3
because this is the slightly more
technically complicated
uh version of the figure three like it
it
figure six basically tells you how you
uh
i mean effectively take a bunch of
states and
like add add them to the superordinate
scale but i think it's slightly easier
to explain
using figure three so what we basically
do
is we're going to take a bunch of states
that comprise our system
and then partition it in partition those
states into markov blankets
and then from there um construct
a uh a reconstruct the system at the
superordinate scale on the basis of
these markov blankets
so uh basically in new work
um especially in the particular physics
monograph
uh we speak of this in terms of two
operators
that we use uh g and r
uh g is a grouping operator effectively
and it allows us to construct
uh markov blankets from the base uh

states

uh that we're considering and then r
takes these mark all blankets and then
reduces their dimensionality

so i mean effectively we're going to be
dropping some of the states

uh to construct the states at the scale
above

so we start with a bunch of states and
uh these are all the states that
are of interest uh to us in the system

uh so you know um

these can i mean these can be literally
any states whatsoever

these techniques have been used recently
to partition

fmri data so when when you're dealing
with fmri data

each state is a voxel so a 3d pixel
effectively

and so what we do from there
is you have all of your states and you
have this continuous time series data
typically

and um so what you do is you construct
an adjacency matrix

um so an adjacency matrix is essentially
you're plotting all of your states by
all of your states

such as the diagonal is basically like
you know your degree of self coupling
and what you do is you populate this
matrix

with uh coefficients that express how
tightly

uh this state and that state are coupled
and then so from this adjacency matrix
what you do

is uh you take the jacobian and the hessian of these adjacency matrices so the jacobian and the hessian are basically the same matrix but rather than having the coupling coefficient uh in each entry what you have is the partial derivatives of each entry so effectively this is a very technical and complicated way of saying what we're looking at is the relative rates of change of every variable with respect to every other variable the reason we're doing this is that a zero in these uh you know uh matrices that contain the partial derivatives means that effectively there's no relative change meaning that like you can you can vary one of the variables and the other one doesn't change so this is a way of testing for conditional independence is the short story so basically using these mathematical techniques what you do is you read off the conditional independencies from this coupling matrix this adjacency matrix and from there you're able to rebuild the uh well you're able to construct markov blankets in terms of the states that are independent of each other conditioned on other states so i mean to recap we started with a

bunch of states that just
composed the system that we're
interested in explaining at one scale
we put these states through a grouping
operator
that chunks it into markov blanketed
particles based on the conditional and
tendencies
and so this takes us from the bottom of
the figure to the top
uh top left so what we're left with
after applying the grouping operator
is a bunch of markup blankets
effectively
mark all blanketed particles so
then in order to get to the next scale
what we do is we reduce the
dimensionality
of this uh partitioned set of states
so effectively what we're doing is we're
dropping the internal states
and the fast um stable modes
of interaction at every scale in order
to construct
the states at the level above and this
gets us uh basically
from the uh it's the it's the other half
of the
image basically moving from these uh
these partitioned uh states
uh back to uh a new set of states at the
superordinate scale so
conceptually what we're saying is that
states
at a superordinate scale are literally
constituted by the slow
meta-stable modes of interaction between
states at the level below

so just conceptually this means that you take say an organ like the heart well the heart itself is constituted by the slower modes of interaction among heart cells so for example like you know uh the uh the main compartments of the heart for example are literally constituted by the modes of interaction uh between the cells that are components of those uh of those components effectively of those compartments um yeah so the then the reduction operator we're dropping the internal states because they're effective all of the information that you need to know about the internal states are already is already summarized in the blanket so you drop the internal states and you drop the fast stable modes of oscillation and that gets you from the uh the partitioned system back to uh a set of states at the next scale um so hopefully that explains what's going on but again the the that might sound very technical but the idea uh i think is pretty simple and it's just that we're constructing superordinate uh layers of the system from the slow modes of interaction of components at the subordinate scale very nice explanation maxwell really

thanks a lot

for that yeah super helpful so let's go

over that because i think it's really

critical and i think for some people

it's like wait

weren't they just talking about ecology

so what just happened there how did we

go from the big questions in ecology

to this grouping operator loop that was

just described

and so the main pivot i think has to do

with multi-scale systems

and how we're going to be modeling

across different scales

and so what the question is is

uh again thinking about understanding

and modeling these dynamics of

multi-scale systems

and there's an unprincipled way and this

is a principled not the only principled

way but

a principled way to do this multi-scale

integration so an unprincipled way

would be for example um you use method a

to go from the fast dynamics to the next

level so you have a heart cell simulator

proprietary program heart cell sim 4 and

then

you make just another software pipeline

that connects the outcome of that

simulation to a model of the whole

heart physiology and then you kind of

run the

the results from these two different

models back and forth so maybe it works

maybe

it's functional so no one's saying it

isn't it's just that the way that those

two pieces were joined together
was a bit a priori it was a little bit
just chosen
and it would be better if there was a
way that we could for example
take measurements across levels and use
information that they provide
to constitute our better understanding
of the whole system
so not just throw away the model that
went um
information that doesn't fit into each
little bottleneck of information
but how can we have this full
variational in the um like
constituting variance across multiple
time scale as well as
variational from that physics side which
brings along
all these really positive benefits
related to computability attractability
and relationship with bayesian
statistics and forney factor graphs and
all these kind of cool areas it ends on
bridging
so if there's a hinge with multi-scale
systems having a useful description
then we're in the category of the
physics models like spin graphs and game
of life
and then also at higher levels at the
very least the tools for dealing with
the kinds of things
that those kinds of simulations like the
game of life were invented to describe
so that's sort of the main key point and
so
this is what maxwell has run through and

represented here
is describing just like you could use
model selection techniques
to figure out which multi-level
statistical regression model or
which multiple anova was the most
appropriate or whether to include an
interaction term or not or a
multi-um uh scale model from a bayesian
framework

whichever modeling framework you're
working within hopefully there's a
principled way
to go about doing model selection and
doing model comparison
and so if i can just interject yes uh
carl

carl fristen um calls this the apart
the particular partition and there's a
there's a play on words here there's a
double on the
by particular partition he means
partitioning the system into particles
which are the internal states
of a system plus the markov blanket but
he also means it's a particular
partition as in there are other ways of
doing this

uh there yeah there there's not
necessarily
one unique way of cutting things up
uh and so the partition itself is named
to reflect that fact i thought it was
kind of fun to bring up
yes now let me connect that point right
there about
one particular choice relating to
dimensional reduction techniques and

where

uh degrees of freedom come into play for
the experimenter so in real data sets
you can do

techniques like principal component
analysis and you can do things like
linear discriminant analysis which looks
basically for group separation
as an additive combination of principal
components

so it tries to explain the most
variation with these orthogonal axes
through the data that's principal
components analysis

now there's some data sets where the
first principal component explains 80
and the next one explains five percent
of the variance and has a long tail
now whether you need the first principal
component and you're just going to take
only the first one

or whether you take 99 of the variance
being explained by 500 principal
components where the
explainability of principal component 74

on your netflix consumption doesn't
really map to like a real world
understandable trait it still can reduce
your um uncertainty about a machine
learning model but it doesn't like map
onto something that's actual about the
group anymore may be misaligned with all
these other things

those kinds of analyses are always up to
the experimenter to decide
how many dimensions they want to reduce
to do you want to reduce
down to a two-dimensional space just

upon the data points in the principal component one principle component two just remap onto the two dimensions that explain the most variants linearly or do you want to map into a 500 dimension space that's where a lot of the freedom comes into play so just like there's a particular principle component analysis is presented in a paper but principal component analysis is a general technique a particular system will have a particular decomposition statistically and in this way and that will also reflect degrees of or freedom in modeling it and in the statistics but the framework just like the anova framework goes a little bit beyond and isn't necessarily um about any specific system well one thing to point out is that there's a kind of recursivity at play where i mean what you can actually see what is the best way to partition the system just kind of recursively in that like what you can do is take your system and then write right alternative generative models of how you think the system is actually behaving and then you can use free energy as a metric to see which out of these alternative models of the system's structure is the one that best explains its behavior so there's a meta-bayesian kind of

aspect to this where like
we're we're not only just like modeling
uh the system as if it were
an active inference agent but we can use
generative models to assess
uh which out of the possible ways of
cutting up the states
of carving up the system is the most
appropriate
and again it's using the same metrics uh
so model evidence
it's always model evidence i had a call
with uh carl yesterday and he was saying
well you know
whatever the question the answer is
always multiple evidence
let me connect that's one more niche
idea so we talked about the skilled
intentionality framework and how that's
like
part of you know the niche is the
affordances and that can reflect a lot
of
aspects that are learned or that are
cultural but
the niche can also include mercury
that's influencing you and you can't
taste it or detect it but it's changing
your phenotype or
something like that so again maybe some
niche concepts are more
affordance oriented like the ecological
psychology niche
is more oriented towards action
unsurprisingly however for many
ecologists when you talk about the niche
that means like
how much light is hitting the ground

what temperature is it at night
so that is also reflected by the
realized versus the fundamental niche
because especially when you're talking
about those abiotic factors
you're sometimes talking about oh yeah
rocks that you can knock over but a lot
of times again people just think of
rocks as being in the physical
niche so things can influence and not be
reflected by sensory states
and then all we have is the model so you
have the model
of decision making it doesn't know that
it's being
slowly drifting off into a non-adaptive
state space because of the
cup that has led into it all it is
experiencing
reducing its uncertainty is about its
generative model
of its niche and that's more on the
ecological psychology side
and that can even be extended to like a
molecular level like chemicals that
can't be tasted
they can still influence you so it's not
only about
how the action is enabled in the world
but that is the primacy
when you're coming from an organism or a
system of interest perspective
and that's where you make this initial
like what are the measurements to make
like maxwell started off with
you start with states those are
measurements you could say
we could measure this we could measure

this thing over here we could do it over here

but you got to start with at least the idea that measurements are going to be made

or at least that there are certain distributions or types of measurements that could be made

um one other thing with the grouping operator that you mentioned

that kind of reminded me of click detection of conditional independencies and so the other way of saying that is click detection of

the conditional dependencies because in order to find out which ones are zeros you got to find out which ones are nonzero and then when you think about there's this representation

ability to transform between the matrix and the network

like through the adjacency matrix or more advanced relation

related matrices then you can do click detection

on a matrix so matrix is going to be able to make

a network that has the exact same data in it

whether it's just one zero like edges or not edges or whether it's a weighted edge or a different type of edge

there's such an equivalence between matrices and networks that click detection

on a social network or something like that is going to be also detectable on the matrix form

so that's something to keep in mind when we're talking about like a matrix coming in here but then there's like maybe a network of interactions coming out another side

okay any thoughts or questions that was definitely worth explaining maybe we could now return to six or we could also go to four where there's a few questions from previous i really wouldn't necessarily go sick honestly

uh it's like the most technical slide like in in our papers sounds good so i mean like unless unless the really nitty

nitty nitty nitty gritty gritty gritty gritty detail

is of interest i would just uh i would suggest that

we not and i think figure three captures everything conceptually

uh the figure six really just tells you how you add states to every particular thing so it's just running through the same thing but like okay we're taking this state

and now what do we do with it uh so yeah we'll do 7.3 late night maybe if we need to

but for now i think let's just return to figure four

which we've actually seen in previous weeks and

i just wrote down a few questions one was

i believe alejandra's question from several weeks ago about

what are some differences between
simulating systems like
those that can at least apparently
engage in deep counterfactuals
and morphological systems and so the
question arose like okay but is the cell
really doing
a counterfactual analysis of alternate
cell shapes it could be
in a way that the brain may be doing
some type of modeling of what
behavioral states could be and then so
that's the previous question that we're
definitely going to return to
hear new perspectives on from anyone who
wants to raise their hand
and then the second set of questions is
cool
where does evolution fit into all of
this and so
a little bit more broadly how does this
bayesian perspective play into evolution
ecology
development and learning because this is
a morphogenesis
problem it's a devo problem and we know
that eco
and evo are going to come into play with
devo so
how do we think about all these levels
and how does
figure four make its appearance in this
paper what what is it relevant what does
it show
so any
thought i think that is just about how
morphogenesis and in general higher
order pattern formation

happens under the variational framework
uh so i mean we've discussed this in
previous episodes but
the idea is always essentially the same
it's that uh so what is it to share a
generative model
well in the context of active inference
to share a generative model means that
you share the same beliefs about
the kinds of sensory consequences
that follow you know the effects of
states in the world and in particular
the effects of
my own actions in the world
so the idea is that if we share a
generative model we expect
to perceive the world in the same way
as a consequence of the same kinds of
things
so uh pursue pursuing on that idea
if we have the same generative model
then if i just do my thing
and you just do your thing we're
eventually going to end up
zeroing in on the same overall pattern
because we share the same expectations
about the kinds of things that we should
expect conditions on the same kinds of
states
so if if you have a system that's
equipped with like
state or yeah state dependent
observation profiles so for example if
i'm a heart cell
i should expect to sense other heart
cells and to
you know register blood flow and
whatever

whereas if i'm a brain cell then i
really shouldn't be registering blood
or other heart cells at all so if you
have these kinds of
state-dependent observation uh
distributions
uh then you can start to see how sharing
a generative model would allow
for a bunch of components to zero in on
uh well first of all on a target
configuration and second of all
kind of settle into a mutually coherent
pattern of inference about what is my
role in these higher order
uh patterns that i'm part of
um yeah and so the the idea there i
guess
is the so this is the hor the vertical
stack idea that we discussed uh last
last week uh this the variational
ecology is a horizontal and a vertical
story
a vertical and a horizontal story it's a
vertical story in that you have nested
systems of systems of systems of systems
which is what i
tend to emphasize in a lot of my work
but at every
basically in between every two scales
there's a niche construction story where
uh you know the the the body at large is
a niche for the organs the organs at
large are a niche for the cells
and so on
thanks i'm gonna address the evolution
side and then return to
this idea that maxwell was just talking
about about shared generative models

because that was also very helpful
so how does evolution play into this
well let's imagine this morphogenesis
creature that we're watching
okay it turns out that there's a lot of
parameters in this model that are
detailed in the paper but
it's not like all parameter combinations
lead to the solution on the bottom right
only specific priors and specific
mapping relationships between sensory
and action states
will result in this exact kind of um
light bulb
shaped creature okay you can imagine
there's some arrangement that might
result in a circle there's some
arrangement it might
the state space is large and it might
not fix upon a stable
point or it might do it very rapidly
what cuts through
all of these models what does model
selection evolution
if it turns out that the wind is blowing
from the north to the south
in this image and that this shape it has
worked in the past and it left more
uh you know offspring that believed that
that shape was the morphogenesis
um position to approach you'll just see
that around it doesn't mean it's the
only solution
it just means it is what has existed so
this is sort of an ab initio
or a computational simulation of an
artificial situation
obviously but in the realized world

where

creatures are evolving then you find
that only the organisms that have the
phenotypes that adequately reflect
regularities their niche

are going to be left carrying forward
and so it's kind of like

wow how do the ants know to drag back
this fly from 30 meters in the desert
it's like

it's all they do it's all they've been
doing if they couldn't do it
they'd be doing something else or they
wouldn't exist it's not like you'd see
an ant just struggling and failing to
do it

that would be a very short-lived
strategy so there's wacky stuff that
happens

again when the niche is transiently
misaligned with evolutionary
priors or when an evolutionary prior
tends towards just like accept
everything into the nest you know pull
in threads or something like that but i
think that's really important to keep in
mind that evolution

often does model selection for us to
actually find

the solutions that work bottom right and
the expectation action relationships
top left that relate in morphogenesis
top right

through the extracellular target signal
being the coordinating mechanism in this
morphogenesis example

on the bottom that's development
mediated what appears to be by this

basically monotonic decrease in free energy representing the utility of modeling this system in converting to or converging towards a developmental attractor

using a free energy framework well notice that it spikes initially which is interesting

so all the agents are kind of trying to figure out what their position is and then they settle into like a mutually consistent uh group inference um and you know this work has been recapitulated across several scales uh

enser palacios and colleagues have a cool paper where they do this across three scales so they they have like a micro meso and a macro scale all premised on the same free energy functional and you get these nice layers of like organelles within organs within an organism

cool and let me also go back to one thing you said about the shared generative model so you talked about like the neurons sort of being in their niche you know wrapped in a specific type of glia or receiving certain types of communication at different developmental time periods from different inputs um

there's almost this um not a complete replacement that you can do of having a shared generative model as you mentioned between the cells but it's

like

they're part of a shared generative

model at a higher level

like through the neurons own

expectations

and its own niche it subsumes and it

becomes

a functional part of a larger generative

model brains know things that neurons

don't

and so that is very naturally

accommodated in a multi-scale framework

because at the level of analyzing just

the two cells

you'd find that they act as if they had

a shared generative model

and so it would make sense oh at the

level of the social interaction it's

like two people are having shared gender

models like they both know how the

movie's going to play out or they both

know whose turn it is to speak

but then at a higher level and that's

all happening at the same time

it's also like being part of a shared

generative model

that is the higher level whether that's

thought of as collective computation or

collective cognition or

distributed intelligence whatever

framework it ends up playing out as

the brain knows things that the neuron

doesn't and so the social group

is having dynamics that constrain and

facilitate the individual level

in a way that specifically will evade

the computation or the understanding

of the individual so i think that's kind

of
an interesting consequence of this and
yeah
good i mean yeah to address a question
that
was asked initially about shared
intentionality um
i mean so we have this paper now out in
entropy
uh where we we basically argue that the
free energy principle provides a formal
semantics
and by that we mean that so what it is
really to act
on a generative model is to have an
intentional relation
with uh some features of
an environment that you're inferring
constantly uh
i mean that that's effectively what
having a generative model
allows you to do it allows you to go
from your sensory data
and from your architecture of priors and
likelihoods to
an estimation of what most likely caused
your sensory data and you know
intentionality
kind of seems to come pre-packaged with
that as in like you know
if intentionality means being uh
appropriately receptive
uh or to you know features of an
environment that stimulate you
well then having a generative model just
means being intentional in that sense
and sharing a generative model it means
that you

you're participating in a form of shared intentionality so a shared way of systematically you know relating uh your observations and embodied priors to um like patterns of behavior and so on cool one other thought and this was something that our friend blue was talking about at our event a couple weeks ago was this multi-scale agency concept so the individual agency concept a relational edge agency then the small group and then the community and so we can't necessarily directly experience but what can you tell somebody to participate in at the local level that does engender this higher order adaptive process to play out and so i think it's the space between having a shared generative model and being a shared generative model like we're going to have shared norms so that we're going to take cognitive diversity and help that map our search space for the critical problems that influence us all so we'll have the best search and we'll figure out the outcomes that are going to be reflecting this kind of a distribution that we want to converge toward that's the kind of well specified harnessing of differences

within the niche because you you said
maxwell um

agents that see the world in the same
way so i i know that you didn't mean
that from the overlapping because
literally no one's you know generative
visual model is going to be the exact
same

but there's this question of how much
seeing the same way
versus basically the variation within
the niche in the exact same way right of
course not but it's
it's a goldilocks situation because if
everyone had exactly the same predictive
model

then we would never see literally
differences in opinions or ideas
and so when thinking about the heterodox
or the novel

or the informative the adaptive that
space is really about the mapping of the
individual

you know uh action perception loops in a
sense

onto this collective level and there's
uh individual states that can be healthy
in one context and and unhealthy or
considered unhealthy in another context
and so it's just i think it's just the
beginning of

thinking about multi-scale systems from
this way but then

as we know people's concept about
themselves and their
social world changes their behavior and
in ways that are really
subtle kind of what is water so just

very interesting stuff

cool any other thoughts on this figure

so just to go through the figures in

reverse order we had six which will

again it's a late night spooky special

for another time

five any thoughts on five also looks

like there's a lot of uh

equations on there not needing to unpack

now

four talked about three and then i mean

yeah

five and six for those of you who really

want to drill into the

i mean we could have like a a 7.3 or

something at some point

uh ideally with a few of the more like

technical people to really unpack that

because five and six like really unpack

the math

uh in some detail but i i don't know how

useful it is for like a high-level

conceptual discuss

discussion like we're having right now

so perfect yep

totally agree and sounds good about that

yep well those were the figures does

anyone else want to return to

any slide or want to bring up any other

thoughts or questions like what was

something that was interesting

that they heard about or something that

they didn't expect for the discussion

to introduce or another direction that

they would like to see it go

towards

well early on there were some remarks

about theory of mind and so on

yeah so we have a whole paper on this
that came out
actually it came out in behavioral and
brain sciences so we're very proud of
that
um it's called thinking through other
minds and
yeah it's it's basically like well it
was a target article and there were like
27 or 28 tier commentaries and we wrote
a response
and so if you're interested in how this
fits in with theory of mind
and you know the the debates between
simulation theory
and theory theory and all that well we
have like a paper
length uh treatment of that
um yeah and the i think the
the kind of in a nutshell response is
that
well you can you can both you can have
your cake and eat it too
using the variational framework i mean
you might have noticed that this is sort
of our
argumentative strategy is often to say
well there seems to be this split in the
literature
and actually both sides of the split can
be made sense of
and done justice too by appealing to
this active inference framework
and here as well i mean we we kind of
strike a balance between these kind of
explicit
theory theory mind reading uh accounts
and these more implicit kind of

simulation theory
embodied resonance kind of approaches
again by appealing to our multi-scale
story right
so i mean you know you have uh more
explicit and more implicit
forms of inference about others and uh
the more explicit they become the more
like the more properly you're heading
into theory of mind territory
um but i mean that that could also be a
topic for discussion that'll in a later
podcast
uh but there's there's a whole paper
length treatment though it was my point
if
it's of interest anyway super cool i'll
check it out
cool yeah i think that it will be really
good for the next
two months last two months of 2020
that's eight weeks
so we'll have four papers the ones that
we laid out
we'll cover a few a few uh
old stomping grounds and a few new areas
as well
for eight we'll be going into math a lot
more
and i think in some other social domains
maybe with the scripts paper
and yeah i'm just like looking forward
to these conversations
making our queue of papers for 2021
having special events if there's anyone
who wants to
arrange for like a special interview or
a

session modeling like a mod like a
simulation
approach these could all be kind of fun
types of discussions to have because
it's fun to bring in all these different
angles on it
hear from the medical and hear from the
practitioners
yeah awesome thank you so much again for
the
the really nice uh treatment that
this uh podcast gives to our work uh
my gratitude is overflowing uh i hope
you can
good yeah these really is unpack and
they help a lot and this is what the
game is about just understanding and
learning and
turning it over hearing in different
ways so thanks for participating
i guess this is 7.2 but next time in 8 9
10
11 for the rest of 2020 we'll we'll
finish out strong
with more learning and more exploring
areas that are new to everyone
so send us any comments or questions
that you have that we can unpack live
the follow-up form is in the calendar
invite so if anyone wants to provide any
feedback there that'd be cool
and just uh stay in touch this has been
a great team com so thanks so much
everyone for participating
very cool take care yep peace
thanks a lot